MSc CLOUD & AI PROJECT · 2026

Airspace Monitor

See the sky. Understand the traffic. Ask the airspace.

Airspace Monitor is a live map of the aircraft visible to the OpenSky volunteer network. It turns free, crowdsourced flight data into a real-time picture of the sky — and adds predictions and an AI assistant on top.

Live map

10,000+ aircraft move smoothly between updates, coloured by altitude. Click one to see its flight, route and predicted arrival.

Unusual traffic

Hexagonal map cells show where traffic is busier or quieter than usual for that hour of the week — with an honest caveat when coverage may be the cause.

Arrival board

Predicted landing times, shown as a range, for flights inbound to Vienna, Bratislava and Budapest.

Ask the sky

A plain-English chat that queries the database and shows the SQL behind each answer, plus a short AI briefing every 30 minutes.

10k+	45–90 s	3	30 min
aircraft in one snapshot	refresh between updates	target airports	AI briefing cadence

WHY IT IS INTERESTING

Real data engineering

Two collectors spool every response before upload. Ingest is idempotent, raw data is archived to R2, and the restore path is documented.

Honest ML

Arrival times are judged against two baselines and returned as a range. Unusual traffic is evaluated with synthetic anomalies because there is no ground truth.

Guarded AI

The chat runs read-only SQL against curated views, shows the query, and is tested against a 30-question golden set with an 80% target.

A map you can look at, a platform you can inspect, and predictions you can question.

Free, non-commercial data. The OpenSky attribution is always visible in the app.

Built by Alek Hanušić & Dániel Csorba.

A live map with a data platform, predictions and a guarded AI assistant underneath.

How it works

Four steps, all built as one system:

1 · Collect
Two home PCs ask OpenSky for every visible aircraft every 90 seconds, taking turns. Each response is saved locally first, so a network or server outage only delays data; it does not lose it.

2 · Store
The cloud server stores positions in PostgreSQL with TimescaleDB and PostGIS. Older raw data is archived to Cloud-flare R2 and read with DuckDB when needed.

3 · Understand
Scheduled jobs build flights, detect squawks and go-arounds, score unusual traffic, look up likely destinations and predict arrival times with a LightGBM model.

4 · Show
A React map streams new snapshots over SSE. The AI briefing and chat read curated database views; the chat can only run read-only SQL and always shows the query.

What is inside

Data platform
Python, FastAPI, PostgreSQL 16 + TimescaleDB + PostGIS, Dagster, Grafana, Cloudflare R2. Idempotent ingest, spool-first collectors, data-quality checks and a documented restore path.

Machine learning
A statistical unusual-traffic baseline, a LightGBM arrival-time model with p10/p50/p90 outputs, and a deterministic route lookup. Every model is compared with a simple baseline and has a public model card.

AI layer
Claude Haiku 5.5 writes the briefing from a facts JSON and answers chat questions through a guarded text-to-SQL path: read-only role, 5 s timeout, allow-listed views and functions, 3 tool calls, 200 rows, SQL shown.

What it cannot do (and says so)
- It only sees aircraft heard by OpenSky's volunteer receivers. A receiver outage can look like a traffic drop; low-traffic cells carry a coverage caveat.
- There are no airline schedules or air-traffic-control data. Arrival times are ranges, and far-out estimates are wider.
- Destinations are estimates, not broadcast data. Diversions can appear late.
- It is not a commercial product, a mobile app or a user-account service.

Status
Week 0 of 8. The plan and technical documents are finished; collection starts first so the models have history.

Core: live map, unusual-traffic heatmap, event feed, arrival board, destination estimate, AI briefing and chat, public dashboards.

Stretch, only if time: a smarter destination model, a smarter anomaly model, holding-pattern events, self-hosted map tiles.

Data and credits: The OpenSky Network (live states and flight lists), Zenodo OpenSky COVID-19 flight dataset, OurAirports, OpenFlights, Natural Earth, aviationweather.gov and the Iowa Environmental Mesonet. Basemap © CARTO © OpenStreetMap contributors. OpenSky data is used for non-commercial research and education; attribution is always visible.